Practical 9: Implement Logistic Regression for Binary Classification and Evaluate the Performance

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In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
df = pd.read_csv('Social_Network_Ads.csv')
```

In [3]:
```
df
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

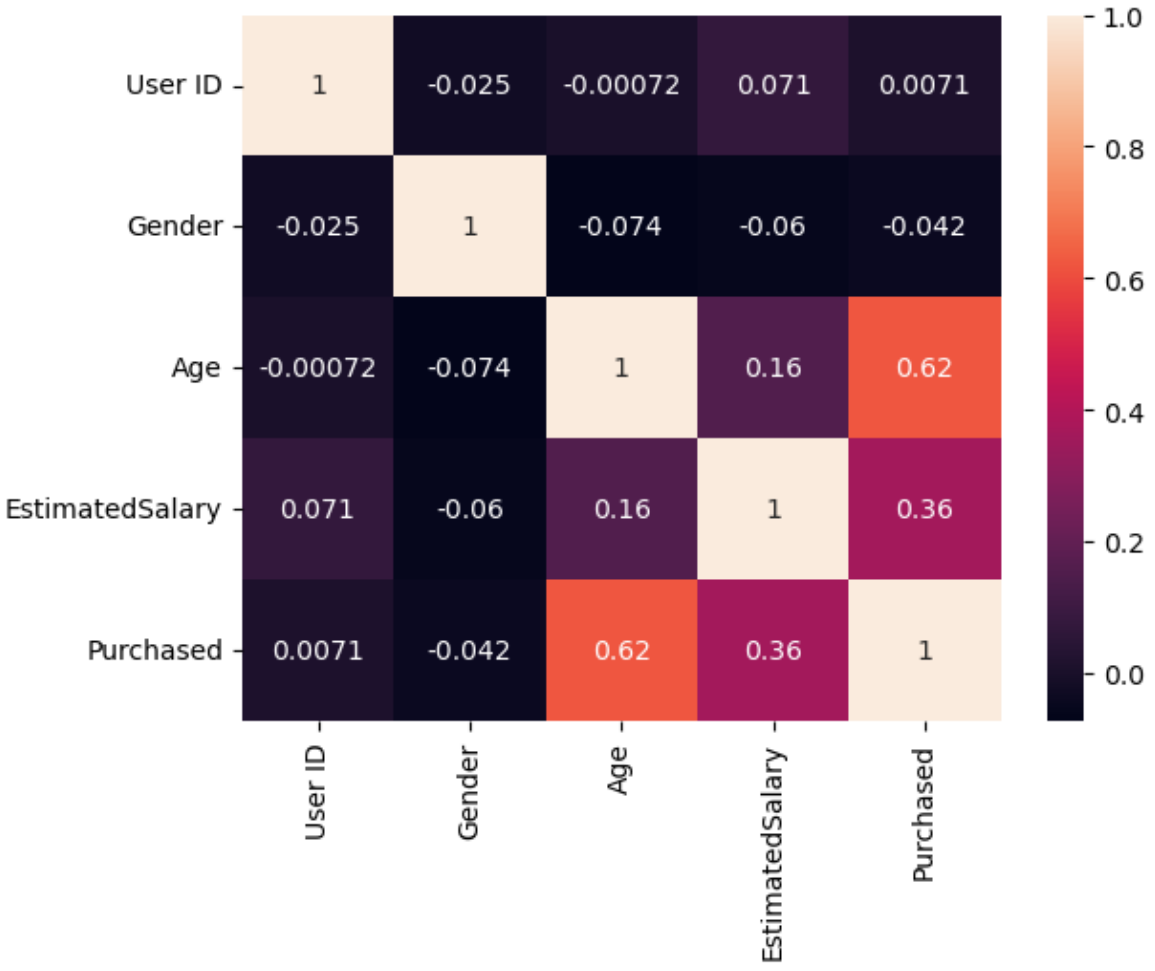
In [4]:
```
df.isnull().sum()
```

Out[4]:
```
User ID            0
Gender             0
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64
```

In [5]:
```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['Gender'] = le.fit_transform(df['Gender'])
```

In [6]:
```
sns.heatmap(df.corr(numeric_only = True),annot = True)
```

Out[6]: <Axes: >



In [7]:
```
X = df.drop(columns = ['Purchased'],axis = 1)
y = df['Purchased']
```

In [8]:
```
X
```

Out[8]:

	User ID	Gender	Age	EstimatedSalary
0	15624510	1	19	19000
1	15810944	1	35	20000
2	15668575	0	26	43000
3	15603246	0	27	57000
4	15804002	1	19	76000
...	...	...	...	...
395	15691863	0	46	41000
396	15706071	1	51	23000
397	15654296	0	50	20000
398	15755018	1	36	33000
399	15594041	0	49	36000

400 rows × 4 columns

In [9]:
```
X
```

Out[9]:

	User ID	Gender	Age	EstimatedSalary
0	15624510	1	19	19000
1	15810944	1	35	20000
2	15668575	0	26	43000
3	15603246	0	27	57000
4	15804002	1	19	76000
...	...	...	...	...
395	15691863	0	46	41000
396	15706071	1	51	23000
397	15654296	0	50	20000
398	15755018	1	36	33000
399	15594041	0	49	36000

400 rows × 4 columns

In [10]:
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, f1_score, classification_repo
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
```

In [11]:
```
from sklearn.model_selection import train_test_split

# Splits features (X) and target (y) into 80% training and 20% testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

In [12]:
```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

In [13]:
```
models = {
    "Logistic Regression": LogisticRegression(),
}
```

In [14]:
```
results = []
```

In [22]:
```
for name, model in models.items():
    model.fit(X_train_scaled, y_train)
    y_pred = model.predict(X_test_scaled)
    acc = accuracy_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    results.append({
        'Model': name,
        'Accuracy': round(acc, 4),
        'F1 Score': round(f1, 4),
    })
```

In [16]:
```
results
```

Out[16]:
```
[{'Model': 'Logistic Regression', 'Accuracy': 0.8125, 'F1 Score': 0.705
9}]
```

In [23]:
```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.81      0.92      0.86        51
           1       0.82      0.62      0.71        29

    accuracy                           0.81        80
   macro avg       0.81      0.77      0.78        80
weighted avg       0.81      0.81      0.81        80
```